THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Substances that taste sour are generally: 1

(a) acidic **(b)** basic **(c)** salty **(d)** neutral

2. A device that converts chemical energy into electrical energy is: 1

(a) an electric cell **(b)** a switch **(c)** a bulb **(d)** a fuse

3. The shiny appearance of metals is called: 1

(a) malleability **(b)** lustre **(c)** sonority **(d)** ductility

4. A change in which no new substance is formed is a: 1

(a) biological change only **(b)** physical change

(c) chemical change **(d)** nuclear change

5. The period of life when a child becomes sexually mature is called: 1

(a) infancy **(b)** childhood **(c)** old age **(d)** puberty

6. Heat flows from a body at higher temperature to a body at: 1

(a) lower temperature **(b)** higher temperature

(c) absolute zero only **(d)** same temperature only

7. The SI unit of time is the: 1

(a) second **(b)** day **(c)** minute **(d)** hour

8. The process of taking food into the body is called: 1

(a) ingestion **(b)** egestion **(c)** digestion **(d)** absorption

9. Plants that make their own food are called: 1

(a) decomposers **(b)** autotrophs **(c)** heterotrophs **(d)** parasites

10. Light travels in: 1

(a) zigzag lines **(b)** circles **(c)** straight lines **(d)** curved lines

11. The Earth spins on its axis from: 1

(a) east to west **(b)** west to east **(c)** south to north **(d)** north to south

12. Which of these is a laboratory acid? 1

(a) lactic acid **(b)** citric acid **(c)** formic acid **(d)** sulphuric acid

13. Materials that allow electric current to pass through them are: 1

(a) fuses **(b)** insulators **(c)** conductors **(d)** cells

14. Non-metals are generally: 1

(a) malleable **(b)** sonorous **(c)** brittle **(d)** ductile

15. Which of the following is a reversible change? 1

(a) melting of wax **(b)** rusting

(c) burning of candle **(d)** cooking egg

16. Ovaries start releasing mature eggs at: 1

(a) birth **(b)** old age only **(c)** puberty **(d)** infancy

17. In conduction, the particles of the solid: 1

(a) move over long distances

(b) fly off

(c) stop

(d) pass on heat to neighbours but stay in position

18. Speed of an object is: 1

(a) total distance **(b)** time per unit distance

(c) total time **(d)** distance travelled per unit time

19. The finger-like projections in the small intestine are called: 1

(a) cilia **(b)** villi **(c)** nephrons **(d)** alveoli

20. Starch stored in leaves turns iodine solution: 1

(a) colourless **(b)** red **(c)** blue-black **(d)** yellow

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. What are indicators? Give two natural indicators. 2

22. What is an electric circuit? 2

23. Name four physical properties of metals. 2

24. What is a physical change? Give two examples. 2

25. What is puberty? At what age does it begin? 2

26. Name the three modes of heat transfer. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Describe how you will test whether a solution is acidic, basic or neutral using litmus paper. 3

28. Explain why the bulb does not glow when the switch is off. 3

29. Describe the activity to test the conduction of electricity in metals and non-metals. 3

30. Classify as physical or chemical change: melting of wax, burning of wood, rusting, boiling of water. 3

31. Explain the changes in the body in adolescents under height, voice and hair. 3

OR

Describe an activity to show conduction in a metal rod. 3

MASTERMIND

32. A pendulum completes 30 oscillations in 60 s. Find its time period and the number of oscillations in 1 minute. 3

OR

Describe the path of food in the human digestive system. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Describe an activity to classify given household substances as acidic, basic or neutral using litmus, turmeric and China rose. 5

34. Describe an activity to show that some materials conduct electricity and others do not. Name five objects. 5

35. Explain the physical properties of metals with an example of each. 5

36. Describe the differences between physical and chemical changes with five examples each. 5

OR

Describe the changes that occur during puberty in boys and girls. 5

37. Explain conduction, convection and radiation with one activity or example each. 5

OR

Describe the experiment to find the time period of a simple pendulum and show how it depends on length. 5

38. Describe the human digestive system and the role of each organ. 5

OR

Describe photosynthesis with the raw materials, conditions, products and the place where it occurs. 5

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND